Hooke two-sphere restitution

LAMMPS dem01 Hooke | Automatic normal damping | 28 October 2026

PASS	All four runs pass the official 2% restitution check and momentum bound at 83.85 us.

Official rebound checks at 83.85 us

Run	Measured e	Error (%)	Left vx (m/s)	Right vx (m/s)	Official
CPU dt	0.4996646896	0.067062	-0.999329379	0.999329379	PASS
GPU dt	0.4996646896	0.067062	-0.999329379	0.999329379	PASS
CPU dt/2	0.4998413098	0.031738	-0.999682620	0.999682620	PASS
GPU dt/2	0.4998413098	0.031738	-0.999682620	0.999682620	PASS

Target e = 0.5; official relative tolerance = 2%. Momentum also satisfies abs(m v_left + m v_right) <= 0.02 (2m) vin = 1.413716694e-05 kg m/s. Both checks apply at step 2795, the end of the second segment. Golden saved e = 0.499664689634 is a separate reference.

Official inputs and FunDEM configuration

Quantity	Unchanged input / direct mapping
Two free spheres	Each R = 0.0025 m; density = 2700 kg/m^3; initial centers x = +/-0.00255 m; vx = +/-2 m/s directed toward contact; zero spin.
Contact law	Hooke kn = 1e6 N/m; restitution e = 0.5. Zero tangential damping and friction in the official linear_nohistory pair command.
Automatic damping	linearStiffnessEnabled = true; dampingEnabled = false turns off manual overrides. Automatic cn = 4.0504686872 N s/m. Explicit coefficient fields are zero and inactive.
Inactive channels	Tangential/rolling/torsional stiffness and friction = 0; zero slip and spin. Automatic tangential damping and active tangent-model equivalence are not tested.
Other settings	No gravity, wall, global damping or constraints. E = 1e4 Pa and nu = 0.33 are inactive schema placeholders under explicit Hooke stiffness.
Time / full output	dt = 30 / 15 ns; 2795 / 5590 steps; same 83.85 us endpoint; output every step including t = 0.

Constant automatic normal coefficient

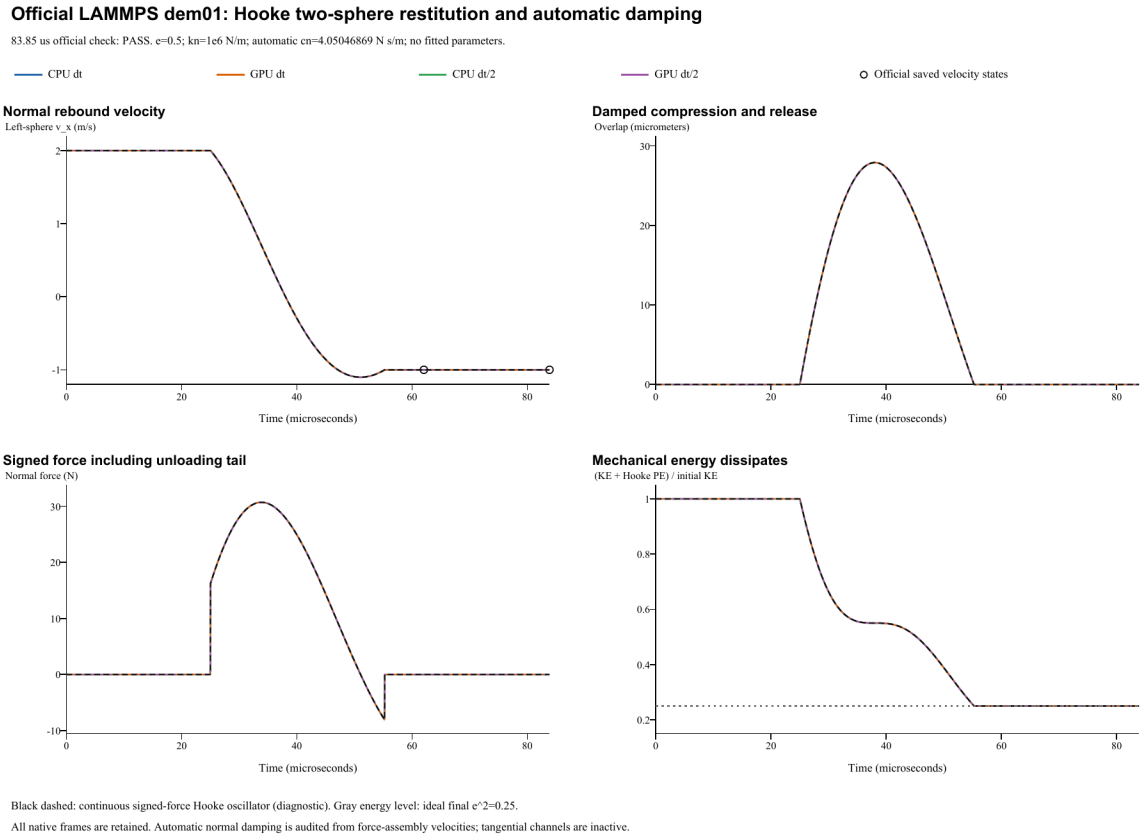
m* = m/2 = 8.8357293382e-05 kg.
beta = -ln(e)/sqrt(pi^2 + ln(e)^2).
cn = 2 beta sqrt(m* kn) = 4.0504686872 N s/m.
Fn = kn delta - cn vn_assembly; signed unloading force is retained.

No parameter fitting, force clipping or contact solver changes. This Hooke case is distinct from the previously tested dem01 Hertz variant.

Reference provenance

Official YAML and saved states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob 57c2041792b5e3744e978e4a1834e46ceaaadb7b. LAMMPS was not rerun locally. YAML epsilon 1e-10 is same-code regression precision, not a universal cross-code tolerance.

Native histories and retained evidence



Automatic coefficient and two-body response

Run	Inferred cn (N s/m)	Min. Fn (N)	Final KE / initial	Max. momentum
CPU dt	4.050468687	-8.104974	0.249664802	0
GPU dt	4.050468687	-8.104974	0.249664802	0
CPU dt/2	4.050468687	-8.092331	0.249841335	0
GPU dt/2	4.050468687	-8.092331	0.249841335	0

Automatic damping reconstructed from native force and force-assembly velocity agrees with the constant Hooke value to a maximum relative error of 7.44e-15; maximum force-reconstruction residual = 7.11e-15 N. Coefficients are inferred diagnostics, not a direct native output channel.

CPU/GPU whole-history maximum velocity / signed-normal-force differences are 4.44e-16 m/s / 7.11e-15 N. Halving dt changes measured CPU restitution by 0.00017662. Restitution improves at dt/2; the sampled peak-overlap error is slightly larger.

Base CPU mechanical energy changes from 0.000706858347058 J to 0.000176477649307 J. The final kinetic-energy fraction is compared with e^2 = 0.25; this is a dissipative collision. Maximum momentum-impulse balance residual across four runs = 2.93e-18 kg m/s.

Inference first removes the final velocity half-kick: vn_assembly = vn_saved - dt Fn/(2m*). Division uses abs(vn_assembly) > 0.08 m/s and delta > 1e-10 m. Signed unloading force is retained. The black dashed oscillator curves and peak compression are additional diagnostics; energy loss is expected.

Complete native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

2796 frames per base-step run; 5591 per half-step run. All 16,774 native frames, contacts, input projects, every-step CSVs, audits and backend logs are retained.

Run manifest | Analysis JSON | Checkpoint CSV | Full native archive ZIP

Execution and scope

Real SimulationSession runs use linked FunDEM 0.4.3 CPU/CUDA native libraries. Installed GUI executable byte identity was not established. Scope: equal-sphere damped normal Hooke impact, restitution, automatic normal damping and momentum. Tangential contact, spin and many-particle flows require separate cases.